

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application No.	19/096,750
Applicant	Dan et al.
Title	Neutralizing Malicious Activities Against Databases
Art Unit	2432
Office Action	Mailed 20 July 2026
Attorney Docket	600276-0002

RESPONSE TO NON-FINAL OFFICE ACTION

Sir or Madam:

In response to the Office Action mailed 20 July 2026, please amend the above-identified application as set forth below and reconsider the application in view of the following remarks. Claims 1-25 are pending. Claims 1, 14 and 20 are amended and claim 26 is added. No new matter is introduced.

AMENDMENTS TO THE CLAIMS

This listing of claims replaces all prior versions and listings of claims in the application. Added text is shown by underlining; deleted text is shown by ~~striketrough~~.

1. (Currently amended) A computer implemented method for protecting a database, the method comprising identifying, by a monitoring system, a malicious activity performed with respect to the database by an incoming connection of a client that is currently running a client SQL statement, wherein the identifying is performed in an agentless manner, without installing a monitoring agent on a server hosting the database; and neutralizing ~~an~~ the identified malicious activity performed with respect to the database while occurring, by dynamically altering a structure of the client SQL statement while the client SQL statement is executing, wherein the altering comprises replacing the client SQL statement with a crafted SQL statement composed in dependence on the identified malicious activity, the crafted SQL statement being executable such that execution of the crafted SQL statement delays operation of the client and holds memory of the client, thereby deterring the client from re-attacking the database.

- 2. (Original)** The method according to claim 1, wherein the database comprises a data masking mechanism and wherein neutralizing the malicious activity utilizes the data masking mechanism.
- 3. (Original)** The method according to claim 1, wherein the malicious activity is generated by a Structured Query Language (SQL) client and comprises an incoming connection currently running a client SQL statement, and wherein the method further comprises altering the structure of the client SQL statement.
- 4. (Original)** The method according to claim 3, wherein altering the structure of the client SQL statement comprises replacing the client SQL statement with a crafted SQL statement.
- 5. (Original)** The method according to claim 3, wherein the altering of the structure of the client SQL statement is performed such that its result would delay the operation of the SQL client.
- 6. (Original)** The method according to claim 5, wherein altering the structure of the client SQL statement comprises utilizing a function configured to pause the operation of the client SQL statement.
- 7. (Original)** The method according to claim 3, wherein the altering of the structure of the SQL statement is performed such that its result would hold significant memory of the SQL client.
- 8. (Original)** The method according to claim 3, wherein the altering of the structure of the client SQL statement is performed in a dynamic manner.
- 9. (Original)** The method according to claim 3, wherein the altering of the structure of the client SQL statement is performed with respect to the client SQL statement and such that the result of the altered SQL statement is related to the client SQL statement.
- 10. (Original)** The method according to claim 3, wherein the database comprises a data masking mechanism and wherein the altering of the structure of the client SQL statement is performed via the data masking mechanism.
- 11. (Original)** The method according to claim 3, wherein altering the client SQL statement comprises: applying a data masking mechanism of the database on the client SQL statement; and executing a crafted function via the data masking mechanism, the crafted function

configured to generate data.

12. (Original) The method according to claim 1, further comprising identifying the malicious activity.

13. (Original) The method according to claim 1, wherein the database is installed on a server, and wherein the neutralizing of the identified malicious activity performed with respect to the database is operated via a monitoring server different from the server.

14. (Currently amended) A system for protecting a database, the system comprising: at least one controller; and at least one non-transitory computer readable storage device storing instructions for execution by the at least one controller, the instructions, when executed, cause the system to identify, in an agentless manner and without a monitoring agent installed on a server hosting the database, a malicious activity performed with respect to the database by an incoming connection of a client currently running a client SQL statement, and to neutralize ~~an~~ the identified malicious activity performed with respect to the database while occurring, by dynamically altering a structure of the client SQL statement while the client SQL statement is executing and replacing the client SQL statement with a crafted SQL statement composed in dependence on the identified malicious activity, the crafted SQL statement being executable such that execution of the crafted SQL statement delays operation of the client and holds memory of the client, wherein the database comprises a data masking mechanism and wherein neutralizing of the malicious activity utilizes the data masking mechanism.

15. (Original) The system according to claim 14, wherein the malicious activity is generated by a Structured Query Language (SQL) client and comprises an incoming connection currently running a client SQL statement, and wherein the instructions, when executed, further cause the system to alter the structure of the client SQL statement by replacing the client SQL statement with a crafted SQL statement.

16. (Original) The system according to claim 15, wherein the altering of the structure of the client SQL statement is performed such that its result would delay the operation of the SQL client.

17. (Original) The system according to claim 15, wherein the altering of the structure of the SQL statement is performed such that its result would hold significant memory of the SQL client.

18. (Original) The system according to claim 15, wherein the altering of the structure of the client SQL statement is performed with respect to the client SQL statement and such that the result of the altered SQL statement is related to the client SQL statement.

19. (Original) The system according to claim 14, wherein the database is installed on a server, and wherein the neutralizing of the identified malicious activity performed with respect to the database is operated via a monitoring server different from the server.

20. (Currently amended) A computer program product comprising a non-transitory computer-readable storage medium having computer-executable instructions for causing the neutralization of an identified malicious activity performed with respect to a database while occurring, wherein the malicious activity is identified in an agentless manner without a monitoring agent installed on a server hosting the database, and wherein the neutralization comprises dynamically altering a structure of a client SQL statement of a client while the client SQL statement is executing and replacing the client SQL statement with a crafted SQL statement composed in dependence on the identified malicious activity, the crafted SQL statement being executable such that execution of the crafted SQL statement delays operation of the client and holds memory of the client.

21. (Original) The computer program product according to claim 20, wherein the database comprises a data masking mechanism and wherein neutralizing the malicious activity utilizes the data masking mechanism.

22. (Original) The computer program product according to claim 20, wherein the malicious activity is generated by a Structured Query Language (SQL) client and comprises an incoming connection currently running a client SQL statement, and wherein the instructions are configured to cause the alteration of the structure of the client SQL statement by replacing the client SQL statement with a crafted SQL statement.

23. (Original) The computer program product according to claim 22, wherein the altering of the structure of the client SQL statement is to be performed such that its result would delay the operation of the SQL client.

24. (Original) The computer program product according to claim 22, wherein the altering of the structure of the SQL statement is to be performed such that its result would hold significant memory of the SQL client.

25. (Original) The computer program product according to claim 22, wherein the altering of the structure of the client SQL statement is to be performed with respect to the client SQL statement and such that the result of the altered SQL statement is related to the client SQL statement.

26. (New) The method according to claim 1, wherein the crafted SQL statement composed for a first occurrence of the malicious activity differs from the crafted SQL statement composed for a subsequent occurrence of the malicious activity.

REMARKS

Reconsideration is respectfully requested. Claims 1, 14 and 20 have been amended to recite the mechanism by which neutralization is achieved, and claim 26 has been added. Support for the amendments appears throughout the specification as filed, including at paragraphs [0067]-[0068] and [0075], and in the passages describing interrogation performed “in an agentless manner, without ‘stepping into’ the servers to be protected.” No new matter is added.

I. Rejection of claims 20-25 under 35 U.S.C. § 101 (transitory media)

Claim 20 has been amended to recite a **non-transitory** computer-readable storage medium, consistent with paragraph [0075] of the specification as filed, which expressly describes a “non-transitory storage medium that may store instructions to perform operations and/or processes.” Withdrawal of this rejection as to claims 20-25 is respectfully requested.

II. Rejection of claims 1-25 under 35 U.S.C. § 101 (abstract idea)

The Office characterizes the claims as reciting a mental process, namely “detecting, analyzing data, data recognition of malicious activity, and manipulating information,” and concludes that the additional elements are generic components implementing a solution “merely at the level of a software.” Applicant respectfully traverses.

A. Step 2A, Prong One: the claims recite no mental process. Amended claim 1 requires that the structure of a client SQL statement be altered *while that statement is executing* on an incoming connection, and that the replacement statement be composed so that its execution delays the client and holds the client’s memory. These operations occur within the execution window of a live database session and act upon a remote client process. They cannot, as a practical matter, be performed in the human mind. A claim that cannot practically be performed mentally does not

recite a mental process. MPEP § 2106.04(a)(2)(III)(C). The rejection does not address the “while occurring” limitation, which is dispositive of the mental-process grouping.

B. Step 2A, Prong Two: any exception is integrated into a practical application. The claimed method does not produce information for evaluation; it produces a changed operational state in the attacking client, namely delayed operation and held memory. That is a concrete technical effect upon a machine, achieved through a specific mechanism, the composition of a crafted, executable SQL statement substituted for the attacker’s statement in flight, and under a specific architectural constraint, namely that no agent is installed on the protected server.

The decisions on which the Office relies are distinguishable on precisely this point. *Electric Power Group* concerned collecting, analyzing and displaying results, with no action taken upon the monitored system. *FairWarning* concerned rules applied to stored audit logs after the fact. *CyberSource* concerned verification that could be performed mentally. In each, the claim ended at analysis. Here the claim begins there and proceeds to act upon the attacking session.

Claims directed to improvements in computer security have repeatedly been held eligible. In *SRI International v. Cisco Systems*, 930 F.3d 1295 (Fed. Cir. 2019), claims to network monitors that detect and respond to suspicious activity were held not abstract. In *Finjan v. Blue Coat Systems*, 879 F.3d 1299 (Fed. Cir. 2018), a behavior-based scanning technique was held eligible as a specific improvement in computer functionality, which rebuts the suggestion that a solution implemented in software is for that reason abstract. See also *Ancora Technologies v. HTC America*, 908 F.3d 1343 (Fed. Cir. 2018); *Enfish v. Microsoft*; *McRO v. Bandai Namco*; *DDR Holdings v. Hotels.com*.

C. Step 2B: the claimed combination is not well-understood, routine or conventional. The record demonstrates the opposite. The closest art of record, Alberstein, neutralizes a suspected injection by converting unknown text elements into non-executable identifiers which, in Alberstein’s own words, “would result in an error when executed rather than performing the malicious functionality.” The claimed method requires the substituted statement to remain executable, and to execute in a manner that consumes the attacker’s time and memory. A claimed combination from which the closest prior art teaches away cannot be conventional.

III. Rejection of claims 1-25 under 35 U.S.C. § 102(a)(1) over Alberstein

Anticipation requires that a single reference disclose each and every limitation of the claim, arranged as in the claim. *Net MoneyIN v. VeriSign*, 545 F.3d 1359 (Fed. Cir. 2008). Alberstein does not.

A. Alberstein’s mechanism is incompatible with the claimed effect. Alberstein is directed to injection detection. It counts risk elements against a whitelist and, where unknown text elements are found, alters them by converting them into non-executable, delimited identifiers. Alberstein states the purpose and the outcome plainly:

“... convert them from potentially malicious instructions into harmless identifiers (which, if malicious, would result in an error when executed rather than performing the malicious functionality).”

A statement that produces an error upon execution terminates. It does not delay the client, does not pause the client’s operation, and does not hold the client’s memory; it releases the resources it was using. Amended claim 1 requires the crafted SQL statement to be executable such that its execution “delays operation of the client and holds memory of the client.” Alberstein therefore does not disclose the claimed limitation, and its disclosed mechanism is incompatible with it.

B. The rejection of claims 5-8 does not identify the corresponding disclosure. The Office maps five distinct limitations to the same passage of Alberstein using materially identical parentheticals. Claim 5 requires that the altering be performed such that its result would delay the SQL client; the cited support states that “an attack would cause delay,” that is, the attack causes the delay rather than the neutralizing step, reversing the causal direction of the limitation. Claim 6 requires a function configured to pause the operation of the client SQL statement; the parenthetical offered is identical to that for claim 5 and identifies no function and no pausing operation. Claim 7 requires that the result hold significant memory of the SQL client; the cited support concerns an attack that would “overwhelm the memory storage allocation,” again describing the attack rather than a defensive statement that occupies the client’s memory. Claim 8 requires that the altering be performed in a dynamic manner; the parenthetical offered repeats the language used for claims 5-7 and does not address dynamism at all, Alberstein’s transformation being a predetermined delimiting operation applied uniformly rather than composed in dependence on the particular activity identified. A statement that a reference discloses a limitation, unsupported by identification of the disclosure relied upon, does not establish a *prima facie* case. MPEP § 2131.

C. Claim 26. Newly added claim 26 requires that the crafted SQL statement composed for one occurrence differ from that composed for a subsequent occurrence. Nothing in Alberstein's fixed delimiting operation discloses or suggests varying the substituted content between occurrences.

IV. Conclusion

Applicant submits that claims 1-26 are in condition for allowance and respectfully requests reconsideration and withdrawal of the rejections. Should any issue remain that might be resolved by telephone, the Examiner is invited to contact the undersigned.

Respectfully submitted,

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